



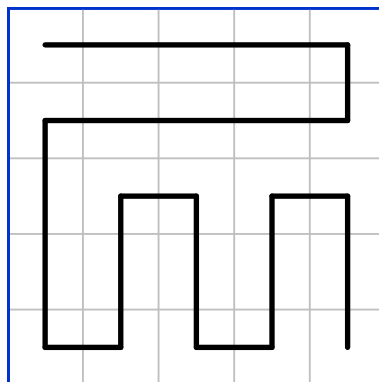
ROUND 1

Instructions and Information

1. Teams can consist of up to 4 people. Each team should sign up before Round 1.
2. This is a 20-question, 2-day test. Each question has a given weight (number of points) shown in the test beside the problem. The maximum number of points you can get is 50.
3. No calculators, books, online resources, etc. are allowed, and no other tab aside from the test paper should be opened. Only blank scratch paper, ruler, compass, and writing materials are allowed. If anyone is caught with illegal resources, they will be disqualified.
4. No person who participated or helped in the contest is allowed to discuss any aspect of it 48 hours from the contest date. If moderators catch anyone discussing during this time frame, they will automatically be disqualified.
5. The cutoff score for Round 2 is the median of all scores.

§1 Problems

1. [1] Three teams, Adelfa, Champaca, and Ilang-ilang play against each other in a tournament. Each team plays against each of the other teams twice. In a given game, the winning team will be awarded five points and the losing team will be awarded zero points. If the game is a draw, each team will be awarded two points. Given that Adelfa has 9 points and Champaca has 7 points, find the sum of all possible number of points that Ilang-ilang has.
2. [1] If both $a + 2b$ and $3b + 4a$ are integers, and $a = \frac{p}{q}$ where p, q are relatively prime positive integers, find the maximum possible value of q .
3. [1] In the game of Chess, a knight attacks any cell that has a horizontal distance of 2 and vertical distance of 1, or a horizontal distance of 1 and vertical distance of 2 from the cell that the knight is on. Now two knights are randomly placed on two cells of a 8×8 chessboard. If the probability that the two knights attack each other can be expressed as $\frac{p}{q}$ where p, q are relatively prime positive integers, find $100p + q$.
4. [1] What is the remainder when 2^{19} is divided by 76?
5. [1] Culver and Kaiser are playing a game. Each of them has a $n \times n$ grid, where n is a random number from 3 to 23, inclusive. Each of them starts at the top left corner of the grid, then tries to trace a path that goes from the top left square to the bottom right square and passes through all the squares. Below is a sample path for $n = 5$:

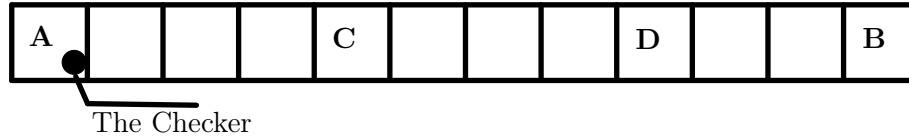


If Culver and Kaiser each get a different value of n , the probability that it is possible for both of them to finish the game can be expressed as $\frac{a}{b}$ where a and b are relatively prime positive integers. Find $a + b$.

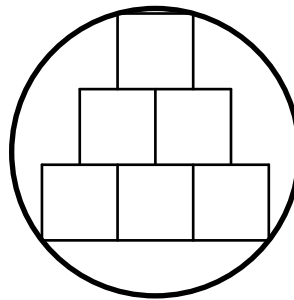
6. [1] Culver has a collection of ten trophies of different heights. He wants to arrange the trophies from left to right, so that for all trophies with two neighbours, it is higher than at least one neighbour. Find the number of ways for him to arrange the trophies.
7. [2] Given that x, y are positive integers satisfying $(\sqrt{x} + \sqrt{x+y}) : \sqrt{y} = 5 : 2$. find the minimum possible value of $x + y$.
8. [2] Let $DSMC$ be a rectangle such that $DS = 4$ and $SM = 3$. Let T be on $\overline{CM}$ such that $\overline{ST} \perp \overline{DM}$, then let H be the intersection of $\overline{ST}$ and $\overline{DM}$. Then, the

area of $\triangle MTH$ can be expressed as $\frac{a}{b}$, where a and b are relatively prime positive integers. Find $a + b$.

9. [2] How many ways can a checker that can move 1, 2, or 3 forward each time, starting at A, ending at B, landing at C at some point and not land at D at any instance?



10. [2] Six squares of side lengths 1 are inscribed in the circle as shown. If the area of the circle can be expressed as $\frac{p}{q}\pi$ where p, q are relatively prime positive integers, find $p + q$.



11. [3] Define $\lfloor x \rfloor$ as the greatest integer not larger than x . Evaluate the sum

$$\sum_{n=1}^{2022} \left\lfloor \frac{1218n}{2023} \right\rfloor$$

12. [3] The rectangular hyperbola with equation $xy = 8$ is rotated through 45° anticlockwise about the origin. Given that the resulting curve can be written in the form $y^2 - x^2 = k^2$, where $k > 0$, find the value of k .
13. [3] In the alphabet puzzle below, where 1 letter represents 1 digit, find the value of ГУУЛБ.

				Г	У	У	Л	Б			
				×	Г	У	У	Л	Б		
				+	У	О	О	О	Л	Б	
				М	Н	О	О	Б	У		
	+	Л	М	Θ	Θ	Г	Б				
		Л	М	Θ	Θ	Г	Б				
		Б	М	У	Л	Г	Б				
		Б	У	Г	О	Б	Б	Б	О	Л	Б

14. [3] An ant moves along the sides of a regular tetrahedron, going from one vertex to another, with each move with equal probability to every other vertices. If the probability that he returns to the original vertex after 6 moves can be expressed as $\frac{p}{q}$ where p, q are relatively prime positive integers, find $p + q$.

15. [3] Let a, b, c be the roots of the equation $2x^3 - 2x + 3 = 0$. The value of $-(a^3 + b^3 + c^3)$ can be expressed as $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$.
16. [4] Count the number of pairs of positive integers (a, b) satisfying $a \leq b < 1000$ and $\frac{\text{lcm}(a, b) + \text{gcd}(a, b)}{a + b} = 3$.
17. [4] Suppose $f(x)$ is a quartic (degree 4) polynomial such that $f(x) \geq x$ for all real x , and $f(-1) = -1, f(0) = 32, f(4) = 4$. Find the value of $f(29)$.
18. [4] Two circles ω_1 and ω_2 are both tangent to line l , but not tangent at the same point. ω_1 and ω_2 are also mutually externally tangent. A circle ω_3 is constructed such that ω_3 is tangent to all of ω_1, ω_2 and l . If the radii of ω_1 and ω_2 are 4 and 9 respectively, and the sum of all possible values of the radius of ω_3 can be expressed as $\frac{p}{q}$ where p, q are relatively prime positive integers, find $p + q$.
19. [4] Let a, b, c be positive integers satisfying $(a - 1)(b - 1)(c - 1) = 2(a + b + c)$. Find the sum of all possible values of $a + b + c$.
20. [5] Acute-angled $\triangle ABC$ has orthocenter H . M is the midpoint of side BC . N is the foot of the altitude dropped from H to AM . If $HN = 8, AM = 27, BC = 36$, and the area of $\triangle ABC$ can be expressed as $\frac{p}{q}$ where p, q are relatively prime positive integers, find $p + q$.

§2 Answers

- | | | |
|--------|-------------|-------------|
| 1. 23 | 8. 443 | 15. 11 |
| 2. 5 | 9. 112 | 16. 232 |
| 3. 112 | 10. 83 | 17. 1125029 |
| 4. 40 | 11. 1230390 | 18. 961 |
| 5. 53 | 12. 4 | 19. 78 |
| 6. 512 | 13. 73325 | 20. 7307 |
| 7. 841 | 14. 304 | |